Notes: Brown, Cookson, and Heimer (JFE, 2019)

Growing up without Finance

Contents

1 Big picture

Question. Does growing up in a place with weak local financial institutions affect how people enter consumer credit markets and manage credit later in life?

Setting. The paper studies Native American reservations. A 1953 federal law, Public Law 280 (PL280), assigned some reservations to state court jurisdiction while others remained under tribal court jurisdiction. Earlier work shows that state-court reservations subsequently developed more local banking activity. Brown, Cookson, and Heimer use this institutional contrast to study whether formative exposure to banking affects household finance outcomes.

Core empirical idea. Tribal-court reservations are treated as relatively financially underdeveloped areas. State-court reservations are treated as relatively more financially developed areas. The main comparison asks whether young adults whose first credit record is on a tribal-court reservation have different credit-market outcomes than comparable young adults from state-court reservations.

Main contribution. The paper shifts the financial development question from firms and local credit supply to household formation of financial behavior. It argues that local financial institutions matter not only because they provide credit today, but also because early exposure teaches people how to use credit and helps create trust in financial institutions.

Main baseline results.

- Young people from tribal-court reservations are much less likely to have a credit report: the median coverage rate is about 53% versus about 73% on state-court reservations.
- Conditional on eventually entering credit markets, they enter more slowly. The probability of receiving a first line of credit is lower under tribal courts.
- In adulthood, tribal-court borrowers have roughly 7–10 point lower Equifax riskscores and about 2–4 percentage point more delinquent accounts, even after controlling for local income and fixed effects.
- These gaps are economically meaningful: the riskscore gap is comparable to a large change in local income and can affect mortgage eligibility and borrowing costs.

Mechanism results. The paper links the credit outcomes to financial literacy and trust. Financial education mandates mitigate the negative effect of growing up without finance. CFPB complaints suggest weaker understanding of credit information. Survey evidence shows that individuals who grew up near a bank have better bank-specific financial literacy and greater trust in bank officials.

Economic takeaway. Local banking markets have persistent consumer-side effects. Weak local finance during formative years delays credit-market entry, worsens credit management, and leaves scars that fade only slowly after people move to more financially developed areas.

2 Data and measurement: key objects

2.1 Reservation institutions and PL280

Native American reservations normally retain substantial legal autonomy, including tribal courts. PL280 changed this institutional environment for a subset of reservations by giving state courts jurisdiction over many civil disputes.

Key institutional split.

- **State-court reservations:** civil contract enforcement is handled by state courts.
- **Tribal-court reservations:** civil contract enforcement remains under tribal courts.

Interpretation. State-court jurisdiction made contract enforcement more predictable for local secured lending. Over subsequent decades, these reservations experienced more local bank development. Tribal-court status therefore proxies for weaker formative exposure to local finance.

2.2 Why the setting is useful

The empirical setting is useful because the institutional difference predates the young borrowers in the sample. PL280 was passed in 1953, local financial development gaps emerged over later decades, and the borrowers in the main sample enter credit markets during 1999–2015.

The lag is central. The law affects the financial environment in which later cohorts grow up, rather than mechanically changing their credit outcomes at the moment of the law.

2.3 Linking consumers to reservation areas

The paper links reservation boundaries to census tracts using TIGER/Line American Indian/Alaska Native/Native Hawaiian Census geographic files. The main geographic unit is the census tract.

Sample geography. The main reservation sample contains 367 reservation census tracts, of which 67 are under state legal jurisdiction and 300 are under tribal courts.

Interpretation. Census-tract geocoding allows the authors to identify people whose first credit report appears on reservation land and to connect those people to the legal-financial environment where they likely grew up.

2.4 Main credit bureau data: FRBNY–CCP

The core data come from the Federal Reserve Bank of New York Consumer Credit Panel (FRBNY–CCP), a 5% random sample of Equifax consumer credit records.

Coverage.

- Quarterly data from 1999Q1 through 2015Q2.
- Consumer-level credit records with census-tract location.
- Rich information on credit reports, credit lines, riskscores, delinquencies, and consumer movement across locations.

Limitation. The FRBNY–CCP contains age but not most demographics, such as race, sex, or education. The authors address this by using geography, fixed effects, census income controls, and complementary survey evidence.

2.5 Main analysis samples

The paper uses three important samples.

Young entry sample. Consumers who were 18 or younger in 1999 and whose first credit report is on reservation land. This sample is used to measure entry into formal credit markets.

Adult stayers. Consumers age 25 or older who remain on reservation land throughout the sample.

This sample is used to measure long-run financial health while holding current reservation residence fixed.

Movers. Consumers whose first credit report is on a reservation but who later move away. This sample is used to examine whether poor formative exposure fades after moving to a more developed financial environment.

2.6 Key outcome variables

Credit report coverage. The fraction of young residents in a tract who appear in the FRBNY–CCP, scaled by 20 because the panel is a 5% sample.

Time to first credit report or first line of credit. Measures how quickly individuals formally enter consumer credit markets after turning 18.

Equifax riskscore. A creditworthiness score similar in spirit to a FICO score. Higher values imply better credit health and cheaper or easier access to finance.

Fraction of accounts delinquent. The number of credit accounts at least 90 days past due, in collections, or severe derogatory, divided by total credit accounts.

Interpretation. Riskscores combine many aspects of credit history. Delinquencies are especially important because they condition on having obtained credit, so they help distinguish credit management from pure access to credit.

2.7 Complementary data: CFPB complaints and Qualtrics survey

CFPB complaint data. Consumer complaints are matched to reservation ZIP codes. The paper studies whether complaints are about credit-information problems and whether the consumer receives relief.

Qualtrics survey. The authors survey Native American respondents age 40 or younger who grew up on or near reservation lands. The survey measures credit-card use, age at first credit card, financial literacy, trust in bank officials, general trust, income, and education.

Interpretation. The survey is not the main identification device. It is used to understand mechanisms: whether growing up near a bank is associated with financial knowledge and trust that are directly relevant to using consumer credit.

2.8 Descriptive facts

In the credit bureau data, the full consumer-quarter sample has 350,798 observations, an average riskscore of 635.7, and an average delinquent-account fraction of 0.14. About 77% of observations are linked to tribal-court reservations, and about 51% are off-reservation observations.

For adult stayers age 25 or older, the mean riskscore is 624.1 and the average delinquent-account fraction is 0.24. In the Qualtrics survey, 66.6% report growing up near a bank, 38.1% have had a credit card, and the average financial trust score is 3.268 on the survey scale.

3 Models and empirical specifications: all equations plus interpretation

3.1 General reduced-form framework

The paper's general empirical comparison can be written as

$$Y_{it} = \gamma_t + \gamma_{\text{census region}} + \beta_1 \text{TribalCourt}_i + \beta_2 \text{Income}_i + \varepsilon_{it}. \tag{1}$$

Objects.

- Y_{it} is a consumer credit outcome for individual i in quarter t.
- TribalCourt_i equals one if consumer i first appears on a tribal-court reservation.
- Income_i is census-tract median household income, measured in 2000.
- γ_t absorbs aggregate time shocks.
- $\gamma_{\text{census region}}$ compares reservations within broad regions.

Interpretation. β_1 is the reduced-form effect of growing up in a reservation environment with weaker local financial development. It is not literally the effect of one additional bank branch. It is the effect of the institutional-financial environment associated with tribal-court status.

3.2 Entry into formal credit markets

To estimate whether consumers enter credit markets more slowly, the paper uses a linear probability model similar to a hazard model:

$$\text{FirstCredit}_{it} = \gamma_t + \gamma_{\text{census region}} + \gamma_{\text{age}} + \beta_1 \text{TribalCourt}_i + \beta_3 \text{Income}_i + \varepsilon_{it}. \tag{2}$$

Dependent variable. FirstCredit_{it} equals one in the first quarter that consumer i receives a first credit report or a first line of credit and equals zero in all preceding quarters after age 18.

Interpretation. A negative β_1 means that consumers from tribal-court reservations are less likely to enter the formal credit system at a given age, conditional on not yet having entered.

Estimation choice. The authors use a linear probability model because later specifications include interaction terms, which are easier to interpret in linear models than in nonlinear hazard models.

3.3 Survey evidence on credit-card use

The survey-based credit-card specification is

$$\text{CreditCard}_i = \beta_1 \text{GrowUpNearBank}_i + \beta_2 \text{Controls}_i + \varepsilon_i. \tag{3}$$

Dependent variables. The outcome is either an indicator for ever having had a credit card or the age at which the respondent first received a credit card.

Key explanatory variable. GrowUpNearBank_i equals one if the respondent reports that there was a bank in the community where they grew up.

Interpretation. This specification directly asks whether formative proximity to a bank predicts consumer credit engagement, holding fixed age, gender, census region, income, and education.

3.4 Adult financial health

For adult credit outcomes, the main specification is

$$Y_{it} = \gamma_t + \gamma_{\text{census region}} + \gamma_{\text{birthyear}} + \gamma_{\text{age}} + \beta_1 \text{TribalCourt}_i + \beta_2 \text{Income}_i + \varepsilon_{it}. \tag{4}$$

Dependent variables. Y_{it} is either the Equifax riskscore or the fraction of accounts delinquent.

Fixed effects.

- Quarter fixed effects absorb aggregate credit-cycle conditions.
- Birth-year fixed effects absorb cohort differences.
- Age fixed effects absorb life-cycle credit patterns.
- Region or region-by-quarter fixed effects absorb broad geographic trends.

Interpretation. A negative β_1 in the riskscore regression and a positive β_1 in the delinquency regression mean that growing up in a weaker local financial environment predicts worse adult credit health.

3.5 Cohort exposure and placebo cohorts

The paper uses older birth cohorts as a placebo. Individuals born before PL280's passage grew up before the major local financial development differences emerged. If the estimates are really about formative exposure, the tribal-court gap should be absent or much weaker for these cohorts.

The paper also allows the effect of tribal-court status to vary by age cohort:

$$\text{RiskScore}_{it} = \gamma_t + \gamma_{\text{birthyear}} + \gamma_{\text{age}} + \beta_1 \text{TribalCourt}_i + \sum_k \beta_{2k} D_k^{\text{age}} + \sum_k \beta_{3k} D_k^{\text{age}} \times \text{TribalCourt}_i + \varepsilon_{it}. \tag{5}$$

Interpretation. The interaction coefficients trace whether the tribal-court credit-score gap is larger for younger cohorts that had more formative exposure to the post-PL280 financial development differences.

3.6 Moving away from reservations

To study persistence, the paper examines people who move away from reservation areas:

$$\begin{aligned} Y_{it} = {} & \gamma_t + \gamma_{\text{birthyear}} + \gamma_{\text{age}} + \gamma_{\text{age when move}} + \gamma_{\text{state move to}} + \gamma_{\text{first census tract}} && (6)\\ & + \beta_1 \text{YearsAway}_{it} + \beta_2 \text{YearsAway}_{it} \times \text{TribalCourt}_i + \beta_3 \text{Income}_{it} && (7)\\ & + \beta_4 \text{RiskScore}_{\text{when move},i} + \beta_5 \text{YearsAway}_{it} \times \text{RiskScore}_{\text{when move},i} + \varepsilon_{it}. && (8) \end{aligned}$$

Objects.

- YearsAway_{it} is the number of years since consumer i moved off the reservation.
- $\gamma_{\text{first census tract}}$ compares consumers who began in the same reservation census tract.
- $\gamma_{\text{state move to}}$ compares movers going to the same destination state.
- The riskscore-at-move controls account for the fact that movers from different reservations may have different initial credit quality.

Interpretation. β_2 measures whether credit health improves faster after leaving for consumers from tribal-court reservations. A positive β_2 in riskscores means convergence; a negative estimate in delinquencies means delinquency rates fall faster.

3.7 Mechanism specifications

The mechanism tests use three complementary designs.

Financial education interactions. The paper augments the entry model with interactions between tribal-court status and state financial literacy mandates, economics mandates, and math reforms. If formal financial education substitutes for early exposure to banks, the interaction with financial literacy mandates should offset the tribal-court disadvantage.

CFPB complaints. The paper estimates

$$\text{CreditOutcome}_i = \beta_1 \text{TribalCourt}_i + \beta_2 \text{Controls}_i + \varepsilon_i, \tag{9}$$

where the dependent variable is either a credit-information complaint or no relief conditional on a credit-information complaint.

Survey literacy and trust. The survey mechanism regression is

$$\text{PersonalAttribute}_i = \beta_1 \text{GrowUpNearBank}_i + \beta_2 \text{Controls}_i + \varepsilon_i. \tag{10}$$

The outcomes include bank-centric financial literacy, stock/bond literacy, financial trust, and general trust.

Interpretation. The mechanism evidence is strongest if growing up near banks predicts bank-specific knowledge and trust in banks, but not unrelated knowledge or generalized trust.

3.8 Standard errors

For the credit bureau specifications, standard errors are generally clustered by census tract and date. For the survey and complaint specifications, standard errors are clustered by state.

4 Identification and credibility

4.1 What identifies the main coefficient

The main coefficient is identified by comparing consumers whose first credit report is on tribal-court reservations with consumers whose first credit report is on state-court reservations, while controlling for income, age, cohort, time, and geography.

The key identifying assumption is that, absent the different local financial development caused by PL280-related court assignment, young people from tribal- and state-court reservations would have had similar trajectories of credit-market entry and credit management.

4.2 Why PL280 is credible for this question

PL280 is useful because the institutional shock occurred decades before the main consumer outcomes are observed. The law created long-run differences in local financial development, but the borrowers in the main sample were not making credit decisions in 1953.

The historical argument is that PL280 assignment was driven in large part by state constitutional and legal artifacts rather than by reservation-level consumer credit prospects. Previous evidence also shows similar pre-PL280 conditions across state- and tribal-court areas in income, unemployment, and banking activity.

4.3 Why direct legal effects are less likely

A central concern is that PL280 might directly affect consumer credit contracts rather than local financial exposure. The paper argues this is limited because most consumer credit products, especially credit cards, are unsecured and governed by lender contracts. Reservation mortgages are also heavily shaped by federal guarantees, reducing the direct role of reservation court enforcement.

Interpretation. The legal shock primarily matters because it changed the local financial ecosystem, not because it mechanically changed credit-card enforcement for young borrowers.

4.4 Controls and fixed effects

The specifications control for census-tract income and include age, birth-year, quarter, census-region, and sometimes region-by-quarter fixed effects. The moving-away specification is especially demanding because it includes first-census-tract fixed effects and destination-state fixed effects.

Interpretation. These controls make the comparison more local and more cohort-specific. They reduce the concern that the results are driven by broad regional economic trends or life-cycle differences.

4.5 Alternative explanations and how the paper addresses them

Economic opportunity. The authors control for income and show robustness to employment, poverty, education, marriage rates, reservation-land share, and older-cohort credit profiles. Delinquency results also condition on having credit, which helps separate management from access.

Current environment rather than formative exposure. The paper uses pre-PL280 birth cohorts and finds little difference across state- and tribal-court reservations for those older cohorts. It also studies movers and shows that the effects fade slowly after people leave the reservation.

Credit demand. The paper studies delinquent accounts and, in appendix evidence, hard credit inquiries. The pattern is difficult to explain solely with lower demand for credit among tribal-court residents.

General education rather than financial exposure. Financial literacy mandates mitigate the tribal-court disadvantage, whereas economics and math mandates do not. This points to a specifically financial-literacy channel.

General trust or ability. Survey evidence links growing up near banks to bank-specific literacy and trust in bank officials, but not to stock/bond literacy or generalized trust.

4.6 Limitations

The design is not a randomized experiment at the individual level. Tribal-court status is a reservation-level proxy for a bundle of institutional and financial-development differences. The credit bureau data also lack demographic variables beyond age. The paper therefore provides a strong reduced-form case that formative exposure to local finance matters, but it does not separately identify every possible micro-channel by which households learn from local banks.

5 Tables and figures

5.1 Figure 1. Timeline of the empirical design

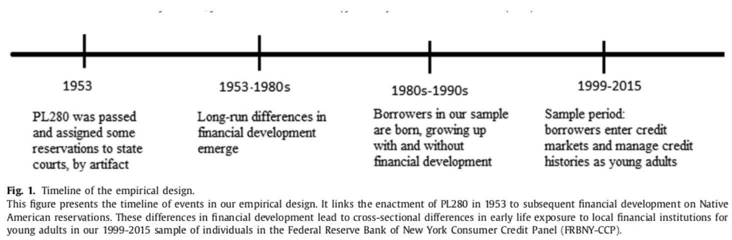


Figure 1: Timeline linking PL280 to later financial development and the 1999–2015 consumer-credit sample.

Read:

- PL280 is passed in 1953.
- Differences in local financial development emerge in later decades.
- The main borrowers grow up after those differences emerge.
- Credit outcomes are observed when they enter and manage credit histories during 1999–2015.

Economic message. The paper is not studying an immediate legal shock to credit reports. It studies how an earlier institutional shock changed the financial environment in which later cohorts formed credit behavior.

5.2 Figure 2. Credit undercoverage across reservations

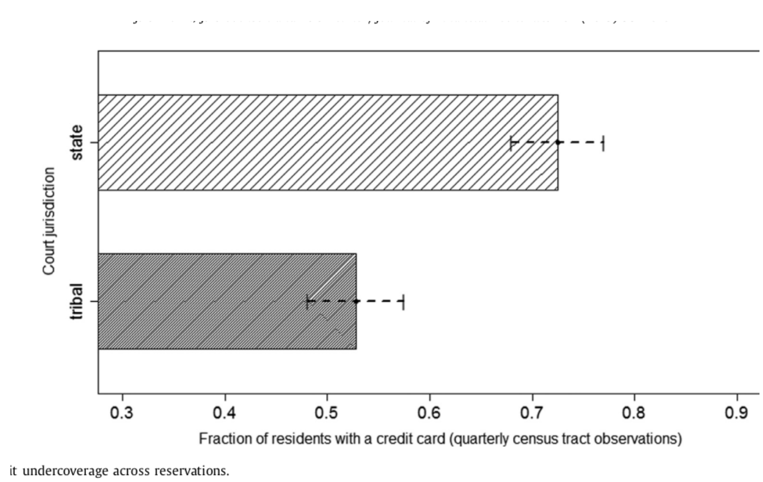


Figure 2: Young residents are less likely to have credit reports on tribal-court reservations.

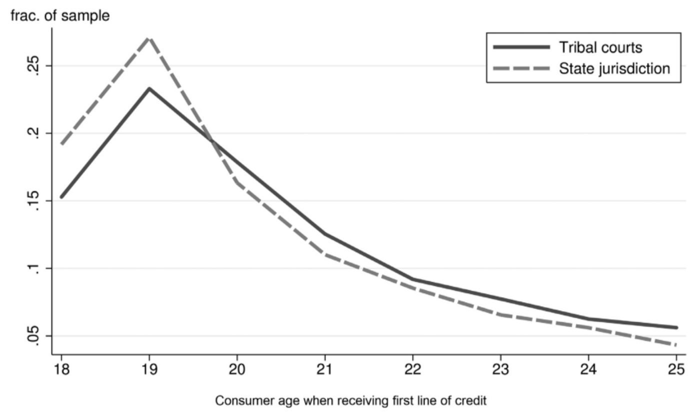
Read:

- Credit report coverage is much lower on tribal-court reservations.

- The median tribal-court tract has roughly 53% young-credit coverage.
- The median state-court tract has roughly 73% young-credit coverage.
- The confidence intervals do not erase the gap.

Economic message. Growing up without local finance delays basic inclusion in the formal credit-reporting system.

5.3 Figure 3. Delayed access to credit



elayed access to credit.
e uses data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes quarterly observations on a panel ner credit records between 1999Q1 and 2015Q2 for borrowers who would have turned 18 by 1999 and whose first credit report is associated with s on Native American reservation lands as defined by the Bureau of Indian Affairs. In state jurisdiction reservations, civil contracts are adjudicated te's court system, as prescribed according to Congressional legislation titled Public Law 280. In tribal court reservations, the tribe's court system es and enforces civil contracts. *Consumer age when receive first line of credit* equals the consumer's age when they obtain their first line of credit.

Figure 3: Age at first line of credit by reservation jurisdiction.

Read:

- State-court borrowers enter credit markets earlier.
- A larger share of state-court consumers obtain a first credit line by ages 18–19.
- Tribal-court consumers are more likely to enter later.

Economic message. The effect is visible at the beginning of credit-market participation, before long credit histories accumulate.

5.4 Table 2. How long does it take to enter credit markets?

Table 2

How long does it take to enter credit markets? Linear probability estimates.

This table presents estimation results from the linear probability model

$$\textit{first credit}_{it} = \gamma_t + \gamma_{census\,region} + \gamma_{age} + \beta_1 tribalcourt_i + \beta_3 income_i + \epsilon_{it},$$

using data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes credit records between 1999Q1 and 2015Q2 for borrowers whose first credit report is associated with a census tract on reservation lands as defined by the Bureau of Indian Affairs. The dependent variable *first credit* equals one in the first quarter that consumer i receives their first line of credit or first credit report (the regressions exclude observations that occur after i has received their first line of credit or first credit report). *Tribalcourt* equals one if the consumer's first credit report is on a reservation using tribal courts as determined by Public Law 280. *Median income* comes from the 2000 US Census and is at the census tract level. Panel A presents the estimates for the full sample. In Panel B, the data are sorted by the state's time-varying status of deregulation under the Interstate Banking and Branching Efficiency Act of 1994. We call the state deregulated if the state has adopted as of quarter t any of the four bank branching measures described in Rice and Strahan (2010). Standard errors are clustered by census tract and date. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Exposure to banks and credit market entry

	time to first line of credit		time to first credit report	
$t = age - 18$	(1a)	(2a)	(3a)	(4a)
tribalcourt	−0.00601***	−0.00510***	−0.00925***	−0.00378*
	(0.0012)	(0.0014)	(0.0018)	(0.0022)
median income / 1000	0.000651***	0.000644***	0.000688***	0.000697***
	(0.000068)	(0.000067)	(0.000082)	(0.000082)
date-quarter FE	x	x	x	x
age FE	x	x	x	x
census region FE		x		x
mean of dep var	0.046	0.046	0.095	0.095
N (consumer-quarter)	246,735	246,735	151,394	151,394
N (consumers)	14,380	14,380	14,380	14,380
R^2	0.047	0.047	0.12	0.12

Panel B: The role of bank branching expansion

	time to first line of credit			time to first credit report		
IBBEA status:	not deregulated	deregulated	all states	not deregulated	deregulated	all states
$t = age - 18$	(1b)	(2b)	(3b)	(4b)	(5b)	(6b)
tribalcourt	−0.0107**	−0.00551***	−0.0386*	−0.0273***	-0.00341	−0.108**
	(0.0052)	(0.0014)	(0.021)	(0.010)	(0.0023)	(0.042)
tribalcourt × deregulated			0.0400*			0.0976**
			(0.021)			(0.042)
median income / 1000	x	x	x	x	x	x
date-quarter FE	x	x	x	x	x	x
age FE	x	x	x	x	x	x
census region FE		x			x	
state FE			x			x
mean of dep var	0.042	0.046	0.046	0.088	0.096	0.095
N (consumer-quarter)	25,197	221,538	246,735	16,404	134,990	151,394
N (consumers)	1,471	12,958	14,380	1,471	12,935	14,380
R^2	0.039	0.049	0.048	0.14	0.12	0.12

Figure 4: Linear probability estimates for time to first credit line and first credit report.

Read:

- Tribal-court status reduces the probability of receiving a first line of credit.
- The effect remains after controlling for census-region fixed effects.
- Bank-branching deregulation under IBBEA partially offsets the tribal-court disadvantage.
- The results are similar for first credit report, though magnitudes differ across specifications.

Economic message. The entry results are consistent with local bank exposure helping young consumers become visible to formal credit markets.

5.5 Table 3. Survey evidence on credit-card use

Table 3
Exposure to banks and credit card usage, survey evidence.
This table uses data from a survey administered by Qualtrics. The survey was targeted to respondents 40 years old or younger who identify as Native American and grew up on or near a Native American reservation. This table presents OLS estimation results of the following specification

$$creditcard_i = \beta_1 grow\ up\ near\ bank_i + \beta_2 controls_i + \varepsilon_i.$$

The dependent variable in columns 1 and 2 is equal to one if respondents answer yes to the question, "Have you ever had or presently have a credit card?" In columns 3 and 4, the dependent variable equals the age when the respondent first received a credit card. The independent variable of interest, *grow up near bank*, equals one if the respondents answers yes to the question, "Was there a bank in the community where you grew up?" Standard errors are clustered by state. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

dep var:	has a credit card (= 1)		age first credit card	
	(1)	(2)	(3)	(4)
grow up near bank (= 1)	0.110***	0.0727**	−1.262**	−1.346**
	(0.034)	(0.033)	(0.52)	(0.53)
male	0.0116	0.00933	1.215**	1.355**
	(0.036)	(0.034)	(0.49)	(0.58)
age	0.0251	0.0268	1.142**	1.271**
	(0.028)	(0.023)	(0.51)	(0.51)
age squared	−0.000339	−0.000373	−0.0138	−0.0161*
	(0.00051)	(0.00043)	(0.0091)	(0.0091)
Census Region FE	x	x	x	x
Income FE		x		x
Education FE		x		x
Number of respondents	704	704	268	268
R^2	0.031	0.096	0.22	0.25

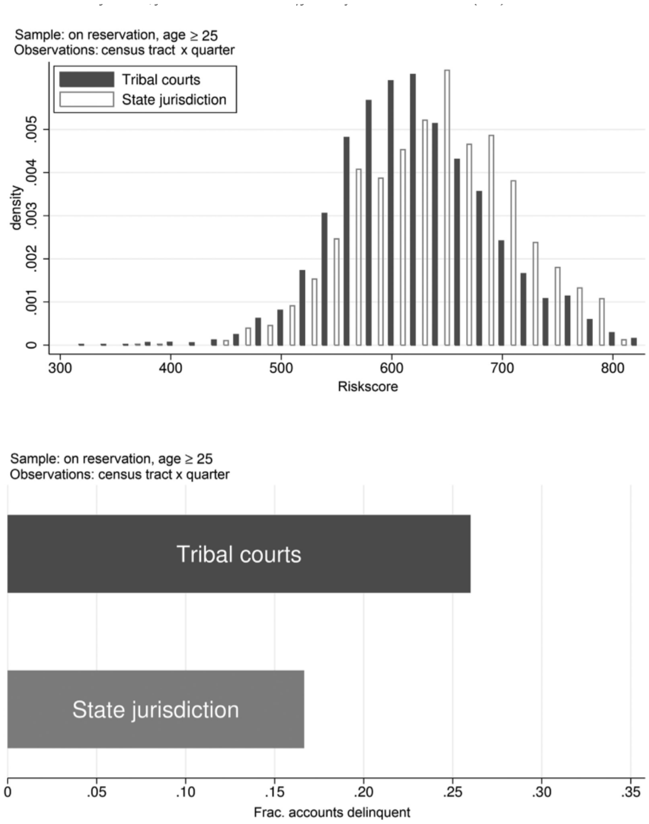
Figure 5: Growing up near a bank predicts greater credit-card use and earlier first credit-card receipt.

Read:

- Respondents who grew up near a bank are 7–11 percentage points more likely to have had a credit card.
- Conditional on having a card, they receive the first card about 1.3 years earlier.
- The result survives controls for income and education.

Economic message. A direct measure of formative bank exposure produces the same qualitative message as the PL280-based design.

5.6 Figure 4 and Table 4. Adult financial health



4. The financial health of reservation borrowers in adulthood—graphical evidence.

Figure 6: Graphical evidence on adult riskscores and delinquent accounts.

Table 4

The financial health of reservation borrowers upon reaching adulthood, consumers 25 years old and older.
This table presents OLS estimation results of the following specification,

$$Y_{it} = \gamma_t + \gamma_{census\,region} + \gamma_{birthyear} + \gamma_{age} + \beta_1 tribalcourt_i + \beta_2 income_i + \epsilon_{it},$$

using data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes quarterly observations on a panel of consumer credit records between 1999Q1 and 2015Q2 for borrowers who turn 18 after 1999, and who are 25 or older as of quarter t. The sample contains consumers whose first credit report has an address on Native American reservation lands and who only appear on reservation lands during the sample. In Panel A, *riskscore* is a proprietary metric from Equifax that measures a consumer's creditworthiness. It ranges from 280 to 850. In Panel B, the dependent variable *frac accounts delinquent* equals the number of accounts at least 90 days past due divided by the total number of credit accounts. *Tribalcourt* equals one if the consumer's first credit report is on a reservation using tribal courts as determined by Public Law 280. *Median income* comes from the 2000 US census and is at the census tract level. Standard errors are clustered by census tract and date. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Financial health measured by riskscore	(1a)	(2a)	(3a)
tribalcourt	−10.14***	−7.521***	−7.035***
	(1.28)	(1.41)	(1.42)
median income / 1000	1.296***	1.219***	1.201***
	(0.035)	(0.041)	(0.041)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	
census region FE		x	
census region × date quarter FE			x
N	45,320	45,320	45,284
R^2	0.070	0.082	0.090
Panel B: Financial health measured by frac. accounts delinquent	(1b)	(2b)	(3b)
tribalcourt	0.0413***	0.0207***	0.0187***
	(0.0061)	(0.0069)	(0.0069)
median income / 1000	−0.00441***	−0.00408***	−0.00406***
	(0.00018)	(0.00019)	(0.00019)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	
census region FE		x	
census region × date quarter FE			x
N	31,795	31,795	31,795
R^2	0.042	0.050	0.060

Figure 7: Regression estimates of adult financial health.

Read:

- Tribal-court borrowers have fewer high credit scores and more subprime scores.
- Delinquent-account rates are higher under tribal courts.
- Regression estimates show a roughly 7–10 point lower riskscore and 2–4 percentage point more delinquent accounts.
- The estimates remain after income, age, birth-year, quarter, and region controls.

Economic message. The formative financial environment affects not only whether people enter credit markets, but also how successfully they manage credit once they have it.

5.7 Figure 5. Robustness to additional controls

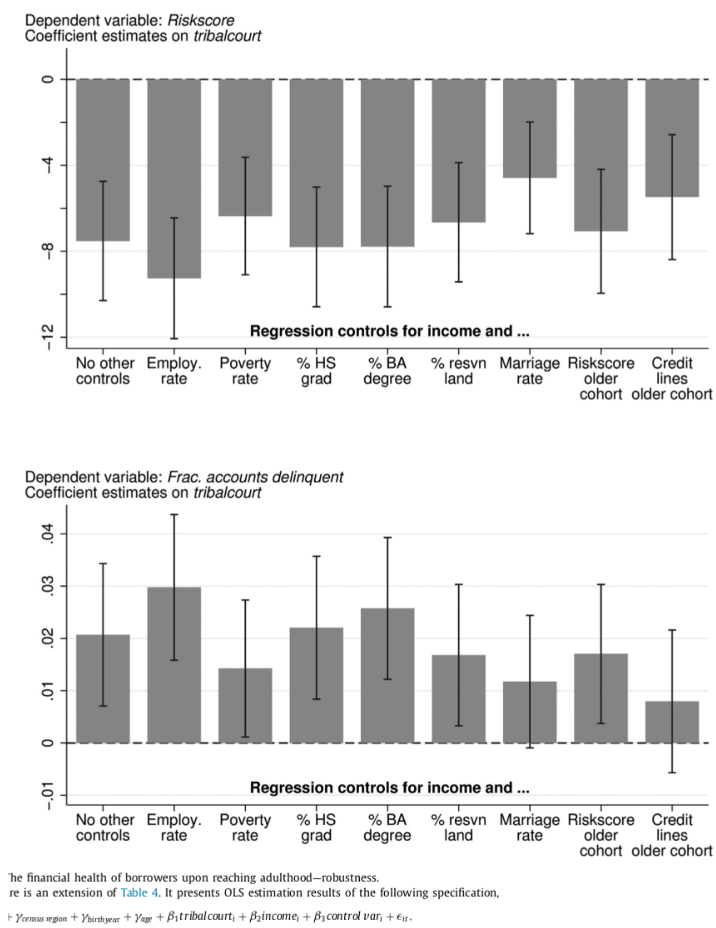


Figure 8: Robustness of the tribal-court coefficient to additional local controls.

Read:

- The riskscore coefficient remains negative under many extra controls.
- The delinquency coefficient remains generally positive.

- Controls include employment, poverty, education, marriage, reservation-land share, and older-cohort credit profiles.

Economic message. The main results are not easily explained by obvious observable differences in local economic opportunity.

5.8 Table 5 and Figure 6. Older cohorts and exposure intensity

Table 5
The financial health of pre-PL280 birth year consumers.
This table presents OLS estimation results of the following specification,

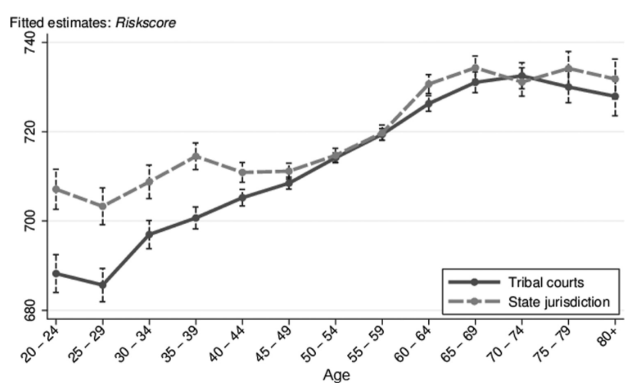
$$Y_{it} = \gamma_t + \gamma_{census\,region} + \gamma_{birthyear} + \gamma_{age} + \beta_1 tribalcourt_i + \beta_2 income_i + \epsilon_{it},$$

using data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes quarterly observations on a panel of consumer credit records between 1999Q1 and 2015Q2 for borrowers born between 1930 and 1953. In Panel A, *riskscore* is a proprietary metric from Equifax that measures a consumer's creditworthiness. It ranges from 280 to 850. In Panel B, the dependent variable *frac accounts delinquent* equals the number of accounts at least 90 days past due divided by the total number of credit accounts. *Tribalcourt* equals one if the consumer's first credit report is on a reservation using tribal courts as determined by Public Law 280. *Median income* comes from the 2000 US census and is at the census tract level. Standard errors are clustered by census tract and date. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Financial health measured by riskscore	(1a)	(2a)	(3a)
tribalcourt	−1.471	3.359	3.317
	(3.93)	(3.81)	(3.81)
median income / 1000	0.941***	0.929***	0.931***
	(0.13)	(0.14)	(0.14)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	
census region FE		x	
census region × date quarter FE			x
N	175,970	175,970	175,970
R^2	0.100	0.120	0.120

Panel B: Financial health measured by frac. accounts delinquent	(1b)	(2b)	(3b)
tribalcourt	0.00355	−0.00287	−0.00284
	(0.0048)	(0.0041)	(0.0041)
median income / 1000	−0.000696***	−0.000703***	−0.000703***
	(0.00015)	(0.00017)	(0.00017)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	
census region FE		x	
census region × date quarter FE			x
N	148,109	148,109	148,109
R^2	0.0085	0.016	0.020

Figure 9: Placebo test using consumers born before PL280's formative effects.



·edit scores by age cohorts.
·e uses data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes quarterly observations on a panel

Figure 10: Credit-score gaps by age cohort.

Read:

- For pre-PL280 birth cohorts, tribal-court status does not predict worse riskscores or higher delinquency.
- The tribal-court riskscore gap is largest among younger cohorts.
- The gap shrinks for older cohorts that had less formative exposure to the post-PL280 banking

differences.

Economic message. These tests are central to the formative-exposure interpretation. Current reservation status alone cannot explain why the gap is concentrated among cohorts exposed during youth.

5.9 Table 6 and Figure 7. Moving away from reservations

Table 6
Moving away from reservations and consumer creditworthiness.
This table presents OLS estimation results of the following specification,

$$Y_{it} = \gamma_t + \gamma_{birthyear} + \gamma_{age} + \gamma_{age\,when\,move} + \gamma_{state\,move\,to} + \gamma_{first\,census\,tract} + \beta_1 years\,away_{it} + \beta_2 years\,away_{it} \cdot tribalcourt_i$$
$$+ \beta_3 income_{it} + \beta_4 riskscore\,when\,move_i + \beta_5 years\,away_{it} \cdot riskscore\,when\,move_i + \varepsilon_{it},$$

using data from FRBNY-CCP, a 5% random panel of consumer credit reports from Equifax. The sample includes quarterly observations on a panel of consumer credit records between 1999Q1 and 2015Q2 for borrowers who have turned 18 by 1999 and whose first credit report is associated with an address on Native American reservation lands as defined by the Bureau of Indian Affairs. In Panel A, *riskscore* is a proprietary metric from Equifax that measures a consumer's credit-worthiness. It ranges from 280 to 850. In Panel B, *frac accounts delinquent* equals the number of accounts at least 90 days past due divided by the total number of credit accounts. *Years away from resvn* is the number of years that have passed since i has moved off of the reservation lands. $\gamma_{first\,census\,tract}$ is a fixed effect for the Census tract of i's first credit report. $\gamma_{age\,when\,move}$ is a fixed effect for i's age when they leave the reservation that is set to zero for all i that do not leave the reservation during the sample. $\gamma_{state\,move\,to}$ is a fixed effect for the state that i moves to when they leave the reservation that is set to zero for all i that do not leave the reservation during the sample. *Tribalcourt* equals one if the consumer's first credit report is on a reservation using tribal courts as determined by Public Law 280. *Median income* comes from the 2000 US Census and is at the census tract level. The regressions also control for i's riskscore when they leave the reservation and its interaction with *years away from resvn*. Standard errors are clustered by census tract and date. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Financial health measured by riskscore	(1a)	(2a)	(3a)
tribalcourt × *years away from resvn*	0.401***	0.411***	0.457***
	(0.086)	(0.087)	(0.087)
years away from resvn	1.987***	1.192***	1.178***
	(0.083)	(0.093)	(0.093)
median income / 1000	0.247***	0.248***	0.250***
	(0.011)	(0.011)	(0.011)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	x
first census tract FE	x	x	x
age when move FE		x	x
state moved to FE			x
N	349,445	349,445	349,445
R^2	0.51	0.51	0.52

Panel B: Financial health measured by frac. accounts delinquent	(1b)	(2b)	(3b)
tribalcourt	−0.00960***	−0.000903***	−0.00113***
	(0.00034)	(0.00033)	(0.00034)
years away from resvn	−0.00742***	−0.000193	−0.0000967
	(0.00036)	(0.00042)	(0.00042)
median income / 1000	−0.000605***	−0.000625***	−0.000600***
	(0.000042)	(0.000044)	(0.000042)
birth year FE	x	x	x
age FE	x	x	x
date quarter FE	x	x	x
first census tract FE	x	x	x
age when move FE		x	x
state moved to FE			x
N	284,722	284,722	284,722
R^2	0.21	0.21	0.22

Figure 11: Moving away and consumer creditworthiness.

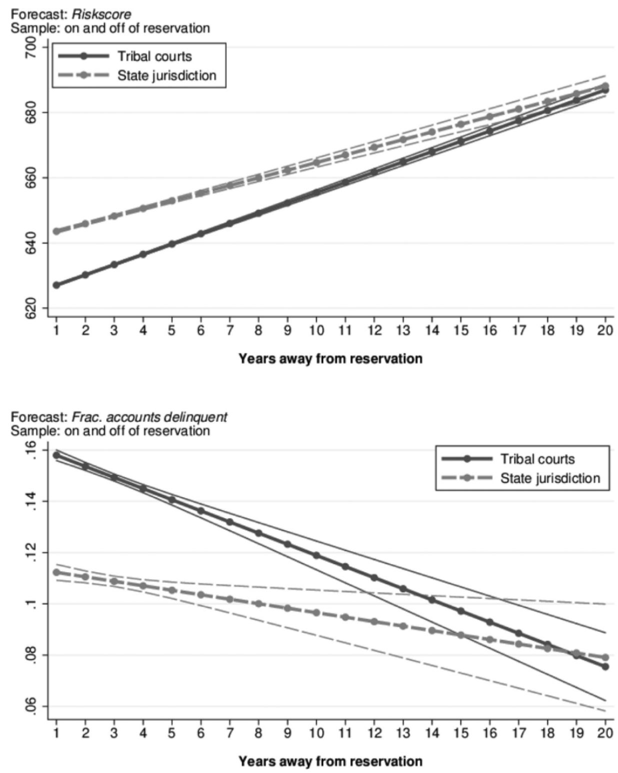


Figure 12: Predicted financial health after moving away from reservations.

Read:

- Credit scores improve after leaving reservation areas.
- Tribal-court movers improve faster than state-court movers, consistent with convergence.
- The initial tribal-court gap is large and closes slowly.
- The figure suggests roughly 17 years for riskscore convergence and roughly 12 years for delinquency convergence.

Economic message. Later exposure to better financial environments helps, but it does not immediately erase the effects of weak formative exposure.

5.10 Table 7. Financial education as a substitute for bank exposure

Table 7
Credit market entry and financial education.
This table presents linear probability estimates using the same samples and specifications as Table 2. This table uses the state-year-level public education reforms from Brown, Grisby, van der Klaauw, Wen, and Zafar (2016). The education variables—financial literacy mandate, economics mandate, and mathematics reform—equal one for consumers in a given state who would have graduated after the new policies were implemented. Standard errors are clustered by census tract and date. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

	time to first credit report			time to first line of credit		
	(1)	(2)	(3)	(4)	(5)	(6)
tribalcourt × financial literacy mandate	0.0429*	0.0401*	0.0460*	0.0243**	0.0252**	0.0239**
	(0.024)	(0.024)	(0.025)	(0.011)	(0.011)	(0.010)
financial literacy mandate	−0.00819	−0.00631	−0.0126	−0.0127	−0.0133	−0.0121
	(0.023)	(0.023)	(0.024)	(0.010)	(0.010)	(0.010)
tribalcourt × economics mandate		0.00772			0.00303	
		(0.013)			(0.0080)	
economics mandate		0.00834			−0.00851	
		(0.014)			(0.0081)	
tribalcourt × mathematics reform			−0.00682			0.00137
			(0.015)			(0.0091)
mathematics reform			0.0120			−0.00195
			(0.015)			(0.0086)
median incomes	x	x	x	x	x	x
date quarter FE	x	x	x	x	x	x
age FE	x	x	x	x	x	x
state FE	x	x	x	x	x	x
N (consumer quarter)	151,394	151,394	151,394	246,735	246,735	246,735
N (conumers)	14,380	14,380	14,380	14,380	14,380	14,380
R^2	0.12	0.12	0.12	0.048	0.048	0.048

Table 8
Consumer complaints and recourse.

Figure 13: Financial education mandates and credit-market entry.

Read:

- Financial literacy mandates offset the tribal-court disadvantage in credit-market entry.
- Economics mandates and math reforms do not produce the same offsetting pattern.
- The effect is visible for both first credit report and first line of credit.

Economic message. The key missing input appears to be specifically financial knowledge, not just general schooling or math ability.

5.11 Table 8. CFPB complaints and credit information

This table uses data from the Consumer Financial Protection Bureau's database of consumer complaints matched to ZIP codes on Native American reservations. It presents OLS estimation results of the following specification,

$$\textit{credit outcome}_i = \beta_1 \textit{tribalcourt}_i + \beta_2 \textit{controls}_i + \varepsilon_i.$$

In columns 1–3, the dependent variable equals one if the complaint claims incorrect information on a credit report. In columns 4–6, the dependent variable equals one if the complaint was resolved without monetary relief. Standard errors are clustered by state. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

sample: dependent variable:	all complaints credit information issue (= 1)			has credit information issue no relief on complaint (= 1)		
	(1)	(2)	(3)	(4)	(5)	(6)
tribalcourt	0.0544***	0.0694***	0.0684***	0.0962*	0.0816*	0.0819*
	(0.015)	(0.018)	(0.021)	(0.056)	(0.043)	(0.047)
personal incomes (ZIP code)			0.00342*			-0.00956
			(0.0018)			(0.0056)
Census region FE		x	x		x	x
Number of complaints	1,604	1,604	1,568	197	197	197
R^2	0.0061	0.013	0.017	0.0083	0.033	0.051

consumer' s complaint is a credit information issue (iden- take directly from Lusardi and Mitchell (2014). The ques-

Figure 14: Consumer complaints and recourse.

Read:

- Tribal-court consumers are more likely to file complaints about credit information.
- Conditional on such a complaint, they are also more likely to receive no monetary relief.
- The result remains with regional and ZIP-code income controls.

Economic message. Consumers from weaker formative financial environments may have less ability to understand or successfully contest credit-report information.

5.12 Table 9 and Figure 8. Financial literacy and trust

Table 9
Financial literacy and trust, survey evidence.
This table uses data from a survey administered by Qualtrics. The survey was targeted to respondents 40 years old or younger who identify as Native American and grew up on or near a Native American reservation. This table presents OLS estimation results of the following specification,

$$\textit{personal attribute}_i = \beta_1 \textit{grow up near bank}_i + \beta_2 \textit{controls}_i + \varepsilon_i.$$

The dependent variable in columns 1 and 2 is equal to the number of correct answers to the following three banking-centric financial literacy questions: “Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have if you left the money to grow?” More than \$102, Exactly \$102, Less than \$102, Do not know, Refuse to answer.’ “Imagine that the interest rate on your savings account by 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?” More than today, Exactly the same, Less than today, Do not know, Refuse to answer. “A 15-year mortgage typically requires higher monthly payments than a 30 year mortgage, but the total interest over the life of the loan will be less.” True, False, Do not know, Refuse to answer. In columns 3 and 4, the dependent variable equals the number of correct answers to the following stock/bond literacy questions: “A 15-year mortgage typically requires higher monthly payments than a 30 year mortgage, but the total interest over the life of the loan will be less.” True, False, Do not know, Refuse to answer. “Buying a single company's stock usually provides safer return than a stock mutual fund.” True, False, Do not know, Refuse to answer. In columns 5 and 6, respondents rate on a scale of 1 to 7, “How much do you trust bank officials to help with your financial decisions?” In columns 7 and 8, the dependent variable equals one if the respondent answers “Most people can be trusted” to the question "Generally speaking, would you say that most people can be trusted or that you have to be very careful in dealing with people?” The independent variable of interest, *grow up near bank*, equals one if the respondents answers yes to the question “Was there a bank in the community where you grew up?” Standard errors are clustered by state. Stars *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

dependent variable:	bank-centric fin. lit.		stock/bond fin. lit.		financial trust		general trust	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
grow up near a bank	0.231***	0.171***	−0.0917	−0.0952	0.596***	0.558***	−0.0127	−0.0245
	(0.065)	(0.061)	(0.073)	(0.076)	(0.18)	(0.20)	(0.024)	(0.027)
male	0.179*	0.165	0.0432	0.0440	0.158	0.115	0.0200	0.0215
	(0.098)	(0.11)	(0.059)	(0.057)	(0.16)	(0.16)	(0.029)	(0.031)
age	−0.0859	−0.0858	0.0296	0.0223	−0.0258	0.00607	0.0361**	0.0322*
	(0.059)	(0.063)	(0.043)	(0.044)	(0.096)	(0.097)	(0.018)	(0.018)
age squared	0.00173	0.00170	−0.000458	−0.000346	0.000422	−0.000111	0.000613*	0.000543*
	(0.0010)	(0.0011)	(0.00079)	(0.00080)	(0.0017)	(0.0017)	(0.00031)	(0.00031)
Census region FE	x	x	x	x	x	x	x	x
Income FE		x		x		x		x
Education FE		x		x		x		x
Number of complaints	704	704	704	704	704	704	661	661
R^2	0.060	0.098	0.026	0.054	0.039	0.068	0.025	0.045

Figure 15: Survey regressions for bank-centric literacy, stock/bond literacy, financial trust, and general trust.

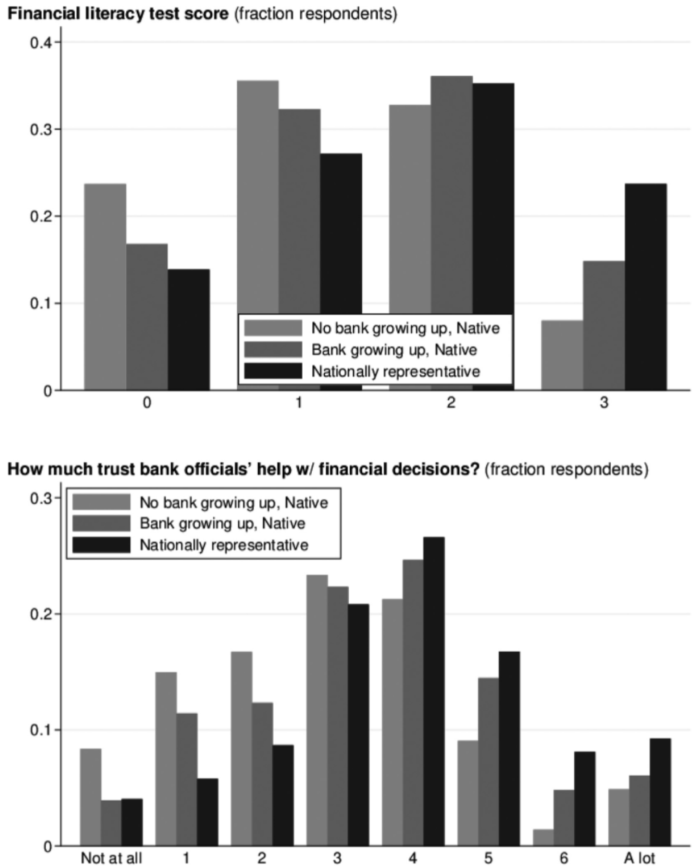


Figure 16: Survey distributions of financial literacy and trust.

Read:

- Growing up near a bank is positively related to bank-centric financial literacy.
- It is not meaningfully related to stock/bond literacy.
- Growing up near a bank is positively related to trust in bank officials.
- It is not related to generalized trust.

Economic message. The mechanism is bank-specific. Exposure to local banking appears to build practical knowledge and institutional trust that are useful for consumer credit behavior.

6 Short summary

- The paper asks whether formative exposure to local financial institutions affects consumer credit behavior over the life cycle.
- It uses Native American reservations and PL280-induced differences in local financial development to compare state-court and tribal-court reservations.
- Tribal-court status proxies for weaker local financial development and weaker early exposure to banking.

- Young adults from tribal-court reservations enter credit markets later and are less likely to have credit reports.
- Adult borrowers from tribal-court reservations have lower credit scores and higher delinquency rates.
- The results are not driven only by current location: pre-PL280 cohorts show little gap, and movers converge only slowly after leaving reservations.
- Mechanism evidence points to bank-specific financial literacy and trust in financial institutions.
- Policy implication: bank access and financial education can have durable effects on household financial health.

7 Conclusion

7.1 Core conclusion: what the paper establishes

The paper establishes that the local financial environment during childhood and young adulthood has persistent effects on consumer credit-market participation and financial health. Individuals who grow up with weaker local banking exposure enter formal credit markets later, accumulate weaker credit profiles, and experience more delinquency.

7.2 Economic intuition

Credit-market participation is partly learned. Young people exposed to nearby banks observe financial products, interact with formal institutions, and build trust in the financial system. Without that exposure, they may delay entry into credit markets, misunderstand credit information, and manage credit less effectively after obtaining it.

7.3 Takeaway

Financial development matters not only because it supplies funds to firms or households today. It also shapes financial human capital. Local bank presence during formative years creates long-lived improvements in financial literacy, trust, credit access, and credit management.